Tutorial 4

MAT3007 Optimization, 2023 Summer

School of Data Science,
Chinese University of Hong Kong, Shen Zhen

6.12-6.16

Outline

1. LP Basics
2. Fundamental LP theory
3. Simplex Method

Review: Basic Terminology

Definition 1 (Polyhedron).

A set P is called Polyhedron if it is the intersection of finite number of linear equations and inequations, which means

$$P = \{x \in R^n \mid Ax \geq b\}$$

Standard form (a special case) would be

$$P = \{x \in R^n \mid Ax = b, x \geq 0\}$$

Definition 2 (Extreme Point).

Let P be a polyhedron. A point $x \in P$ is said to be an extreme point of P if we cannot find two elements $y, z \in P$, both different from x, and a scalar $\lambda \in [0, 1]$, such that $x = \lambda y + (1 - \lambda)z$

Review: Basic Terminology

Definition 3 (Basic Solution).

We call x a basic solution of the LP if and only if
(1) $Ax = b$.
(2) There exist indices $B(1), \ldots, B(m)$ such that the columns of

$$\begin{bmatrix} | & | & & | \\ A_{B(1)} & A_{B(2)} & \cdots & A_{B(m)} \\ | & | & & | \end{bmatrix}$$

are linearly independent and $x_i = 0$ for $i \neq B(1), \ldots, B(m)$.

Definition 4 (Basic Feasible Solution).

x is a basic feasible solution of a LP if and only if x is a basic solution and $x \geq 0$.

Exercise 1

For the standard LP polyhedron $\{x : Ax = b, x \geq 0\}$, the followings are equivalent:

1. x is an extreme point
2. x is a basic feasible solution

Solution to Exercise 1: (1) $\rightarrow$ (2)

Suppose x is an extreme point, $B = \{B(1), B(2), \ldots, B(k)\}$ be the set of indices such that $x_{B(i)} > 0$. Prove $A_{B(1)}, \ldots, A_{B(k)}$ are linearly independent.

Suppose these columns are linearly dependent, which means $\exists \alpha = (\alpha_1, \cdots, \alpha_k)^T \in \mathbb{R}^k$, where α_j 's are not all zeros, such that $\sum_{j=1}^{k} \alpha_j A_{B(j)} = 0$

Now $\forall i \in B$, we can choose a small ϵ such that both $\bar{x}_i = x_i + \epsilon\alpha_i, i \in B$ and $\tilde{x}_i = x_i - \epsilon\alpha_i, i \in B$ are still feasible solutions.

Then we have $x = 0.5\bar{x} + 0.5\tilde{x}$, which contradicts with that x is an extreme point.

Solution to Exercise 1: (2) $\rightarrow$ (1)

Suppose x is a BFS, we want to prove x is also an extreme point.

Suppose x is not an extreme point, then there exists two feasible solutions $x^{(1)}, x^{(2)} \neq x, \lambda \in (0,1)$, such that $x = \lambda x^{(1)} + (1-\lambda)x^{(2)}$.

We then arrange x in the form $(x_B; x_N)$ and see what happens:

$$\begin{pmatrix} x_B \\ x_N \end{pmatrix} = \lambda \begin{pmatrix} x_B^{(1)} \\ x_N^{(1)} \end{pmatrix} + (1-\lambda) \begin{pmatrix} x_B^{(2)} \\ x_N^{(2)} \end{pmatrix}$$

$$x_N = \lambda x_N^{(1)} + (1-\lambda)x_N^{(2)}, x_N = 0, x_N^{(1)} \geq 0, x_N^{(2)} \geq 0 \Rightarrow x_N^{(1)} = x_N^{(2)} = 0$$

$$Ax^{(1)} = Ax^{(2)} = b \Rightarrow x_B^{(1)} = x_B^{(2)} = A_B^{-1} b = x$$

Q.E.D

Outline

1 LP Basics

2 Fundamental LP theory

3 Simplex Method

Review: Fundamental LP Theorem

Consider a linear problem **in standard form** and assume that A has full row rank m.

1. **existence of extreme points**:
   If the feasible set is nonempty, there exists a BFS. (And we know for a standard LP polyhedron, extreme points and BFS are equivalent)
   Remark: $x \geq 0$ plays an important role in the existence!
2. **optimality of extreme points**:
   If optimal solution exists, there is an optimal solution that is a BFS.
   Remark: We only need to look among BFS for an optimal solution. But if the optimal cost is not bounded, we do not have an optimal solution for the LP problem, and the optimal cost can not be attained by an extreme point.

Exercise 2

Consider the standard form polyhedron $P = \{x \in \mathbb{R}^n \mid Ax = b, x \geq 0\}$. Suppose $A \in \mathbb{R}^{m \times n}$ and has m independent rows. True of False?

(a) if $n = m + 1$, then P has at most two basic feasible solutions.
(b) The set of all optimal solutions is bounded.
(c) At every optimal solution, no more than m variables can be positive.
(d) If there is more than one optimal solution, then there are unaccountably many optimal solutions.
(e) If there are several optimal solutions, then there exist at least two optimal basic feasible solutions.
(f) Furthermore, for any arbitrary nonempty polyhedron, it has at least one extreme point.

Solution to Exercise 2

Consider the standard form polyhedron $P = \{x \in \mathbb{R}^n \mid Ax = b, x \geq 0\}$. Suppose $A \in \mathbb{R}^{m \times n}$ and has m independent rows. True of False?

(a) if $n = m + 1$, then P has at most two basic feasible solutions.
✓ n is the dimension of x, $m(= n - 1)$ is the number of equality constraints. Now x lies in a n-dimensional space but $n - 1$ degrees of freedom are taken away, so actually x lies in a line in this n-dimensional space and for any line, it has at most 2 vertexes.
(b) The set of all optimal solutions is bounded.
× Consider the following toy example, $P = \{(x_1, 0) \in R^2 \mid x_1 \geq 0\}$ is the unbounded optimal solution set.

$$\begin{array}{ll} \min & x_2 \\ \text{s.t.} & x_1 \geq 0 \\ & x_2 \geq 0 \end{array}$$

Solution to Exercise 2

(c) At every optimal solution, no more than m variables can be positive.
$\times$ Recall the toy example in (b), $m = 0$, but every point in $S = \{(x_1, 0) \in R^2 \mid x_1 > 0\}$ has 1 positive variable and optimal (they are all not extreme points!).

(d) If there is more than one optimal solution, then there are unaccountably many optimal solutions.
$\checkmark$ By convexity of polyhedron.

Solution to Exercise 2

(e) If there are several optimal solutions, then there exist at least two optimal basic feasible solutions.
$\times$ Same as in (b), $P = \{(x_1, 0) \in R^2 \mid x_1 \geq 0\}$ has several optimal solutions but only $(0, 0)$ is extreme point i.e. BFS.

(f) For any arbitrary nonempty polyhedron, it has at least one extreme point.
$\times$ Choose $P = \{(x_1, x_2) \in R^2 \mid x_1 - x_2 \geq 0\}$ which is a half space, so no vertex (but if we add $x \geq 0$ in P, it has!)

Outline

1 LP Basics

2 Fundamental LP theory

3 Simplex Method

Review: Simplex Procedure

Recap on simplex procedure for standard form LP:

1. Find a BFS with basis B.
2. Find reduced cost $\bar{c}_j$ for j $\notin B$ If $\exists j$ s.t. $\bar{c}_j < 0$, then continue; otherwise stop. ($\rightarrow$ Stopping Criterion)
3. j th direction is $\mathbf{d} = \left(-A_B^{-1}A_j, 0, \cdots, 1, \cdots, 0\right)^T$ If $\mathbf{d} \geq 0$, then unbounded; otherwise for some $d_i < 0$, choose $\theta = \min_{(i \in B | d_i < 0)} \left(-x_i/d_i\right)$
4. $\mathbf{y} = \mathbf{x} + \theta\mathbf{d}$ is a new BFS with basis B'
5. go to (1)

Exercise 3

Use the simplex method to solve the following problem (This trivial problem is an illustration of simplex method.)

$$\begin{aligned} \min \quad & 3x_1 + 4x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4 \\ & x_2 \leq 5 \\ & x \geq 0 \end{aligned}$$

Solution to Exercise 3

Standardization:

$$\begin{aligned} &\min\ 3x_1 + 4x_2 \\ &\text{s.t. } x_1 + x_2 + x_3 = 4 \\ &x_2 + x_4 = 5 \\ &x \geq 0 \end{aligned}$$

$$\mathbf{c} = (3, 4, 0, 0)^T \quad \mathbf{x} = (x_1, x_2, x_3, x_4)^T \quad \mathbf{b} = (4, 5)^T$$

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

Note: A has 2 linearly independent rows. To simplify the solution, we would use $[x_B; x_N]$ to compact several column vectors x_B and x_N rather than $\left(x_B^T, x_N^T\right)^T$.

Solution to Exercise 3

(1) Suppose we choose $B(1) = 1$ and $B(2) = 4$, we re-order all entries by $\mathrm{B} = [B(1)B(2)]$ and $\mathrm{N} = \mathrm{B}^c$

$$A_B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

then, a BFS is $x = [x_B; x_N]$ where $x_B = A_B^{-1}\mathbf{b} = [4; 5]$ and $x_N = 0$
(2) compute $\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$ (we start from the smallest index) $\bar{c}_2 = 1 \quad \bar{c}_3 = -3$. Choose index 3 .
(3) Since $\mathbf{d} = [\mathbf{d}_B; \mathbf{d}_N] = \left[-A_B^{-1}A_j; 0; 1\right] = [-1; 0; 0; 1]$ (index : 1423),
$\theta = \min_{(i \in B | d_i < 0)} (-x_i/d_i) = 4$
(4) $\mathbf{y} = x + \theta d = [0; 5; 0; 4]$ (index: 1423)
(5) we repeat the above (2)-(4) and find out this is optimal. We conclude: optimal solution $[x_1; x_2] = [0; 0]$ and optimal value $= 0$.

Acknowledgements

This slide is based on previous tutorial materials in fall and spring semesters for MAT3007. We thank previous MAT3007 TAs and instructors for sharing previous course materials.